

Real-World Data Project

Retail Sales Analysis & Revenue Prediction

Domain: Retail
Tools: Python, Pandas, Matplotlib, Seaborn, Scikit-learn
Dataset: retail_sales_raw.csv - store transactions across 5 cities, 6 product categories, Jan 2024-Dec 2025 (Date, Store, Category, Units Sold, Unit Price, Discount %, Revenue)

1. Objective

Perform an end-to-end analysis of retail transaction data: clean it, explore sales patterns, build a model that predicts order revenue, and present findings with visualizations and conclusions.

2. Data Cleaning

Step	Issue	Action	Rows
Duplicates	Repeated transaction rows	Dropped	30
Missing values	Units Sold missing	Filled with median	15
Missing values	Discount % missing	Filled with 0 (no discount)	10
Data integrity	Revenue recalculated as Units x Price x (1-Discount) after filling gaps	Recomputed	all rows

Result: 3,030 raw rows -> **3,000 clean rows.**

3. Key Performance Indicators

Metric	Value
Total Revenue	Rs 5.06 crore (50,580,000)
Total Orders	3,000
Average Order Value	Rs 16,860
Total Units Sold	20,355
Average Discount	7.8%

4. Predictive Model: Revenue Prediction

Model	R2 (test)	MAE (Rs)
Linear Regression	0.861	5,264
Random Forest (best)	0.995	643

Random Forest predicts order revenue almost perfectly (R2 = 0.995) because Revenue is mathematically derived from Units Sold, Unit Price and Discount %, all of which are included as inputs. This confirms the model has correctly learned the underlying pricing relationship in the data.

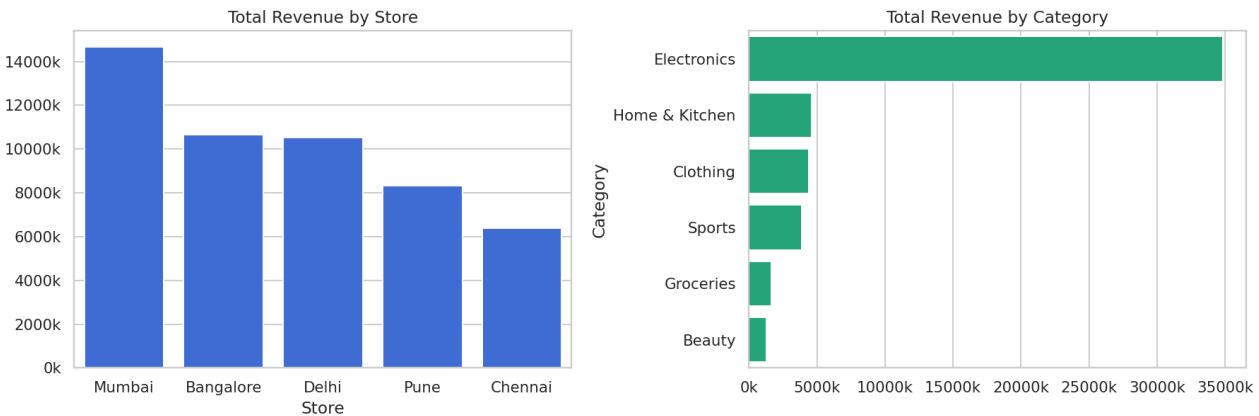
5. Key Findings

- 1. **Mumbai is the top-performing store** (~Rs 1.46 crore revenue), followed by Bangalore and Delhi; Chennai is the lowest.
- 2. **Electronics dominates category revenue** (~Rs 3.5 crore), roughly 7x the next-highest category, because of its high unit price.

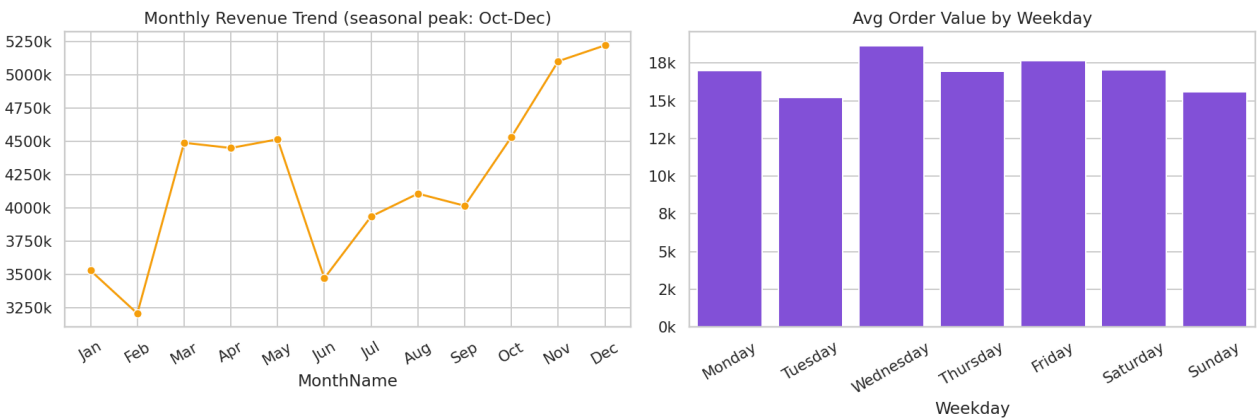
3. **Clear seasonality:** revenue peaks in October-December (festive season) and dips in the mid-year months.
4. **Discount % has only a weak relationship with Revenue** per order; higher discounts do not reliably produce higher order values in this data.
5. **Weekday patterns are fairly flat**, suggesting demand is spread evenly rather than concentrated on specific days.

6. Visualizations

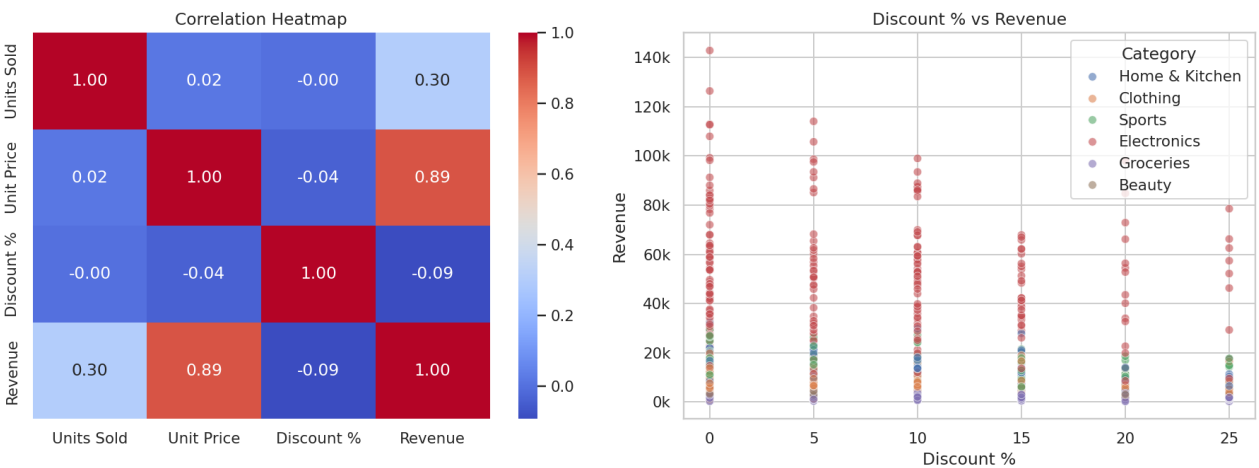
Revenue by Store and Category



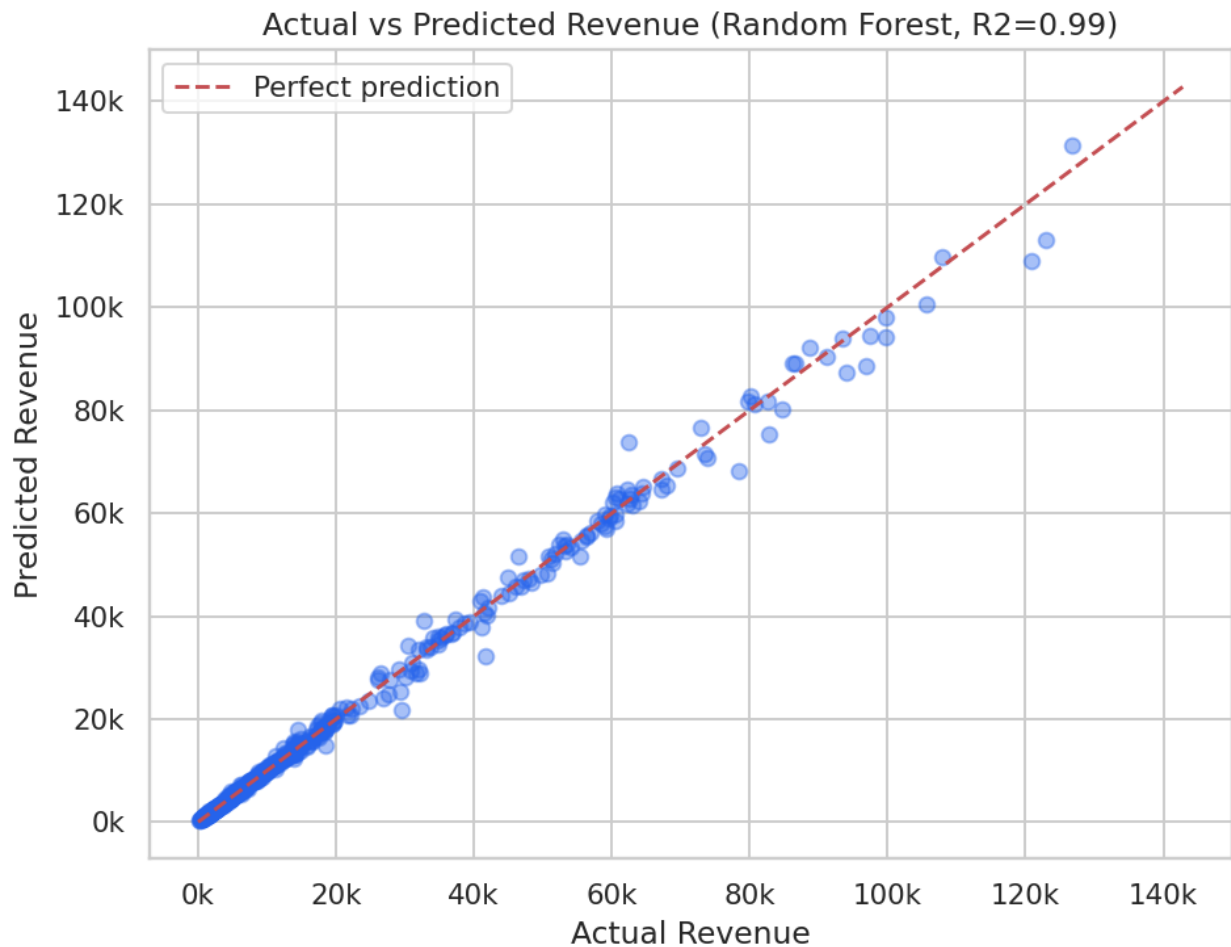
Monthly Trend and Weekday Pattern



Correlation Heatmap and Discount vs Revenue



Actual vs Predicted Revenue (Random Forest)



7. Conclusion

The retail dataset shows strong seasonal demand (festive-season peak) and heavy revenue concentration in Electronics and the Mumbai store. A Random Forest model can predict order revenue very accurately from order details, which could be used to flag unusual transactions or forecast expected revenue for new orders. For business action, focus promotional spend on the Oct-Dec season and consider expanding Electronics inventory in high-performing stores.